

# Kumbara by Sembol

• TESTNET

A self-custodial dollar piggy bank for Türkiye, opened with Face ID.

Live (Stellar TESTNET)

**kumbara.sembol.xyz**

Repository

**github.com/keyboard01/kumbara**

Deck date

**6 September 2026**

# Lira savings lose value; the dollar route has a wall in front of it.

## Inflation makes lira saving a losing move

Consumer prices in Türkiye rose **31.51% year on year** in August 2026.<sup>1</sup> Dollar-denominated saving is how ordinary households defend themselves.

## The crypto route exists but sheds users at every step

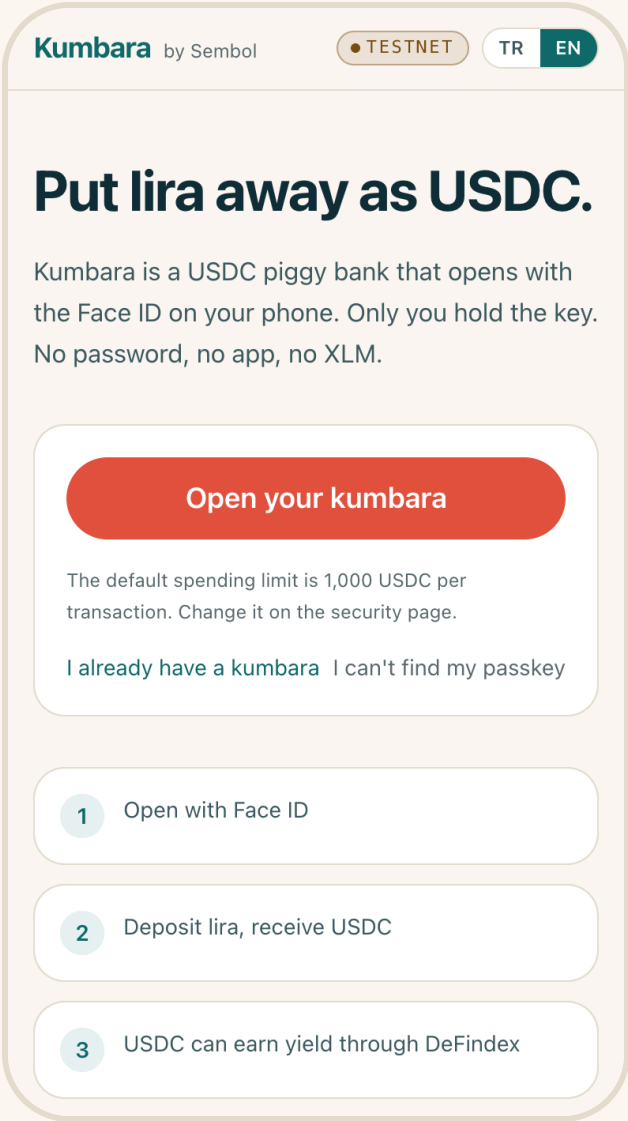
USDC on Stellar settles in seconds for fractions of a cent. Getting there means an extension or app install, twelve words to keep safe, XLM for gas bought somewhere else, and a phone you must never lose.

## The people who need hard-currency savings most are served least

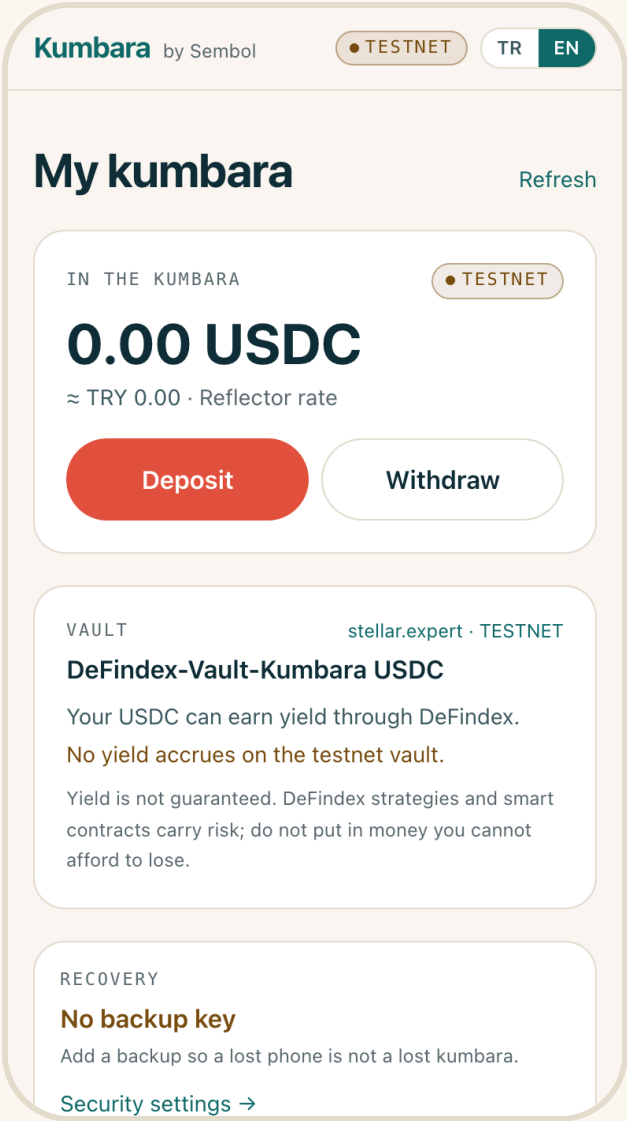
A bank-transfer habit and a phone with Face ID are what people have. That is the whole interface Kumbara asks for: no seed phrase, no XLM, no install.

<sup>1</sup> TÜİK (Turkish Statistical Institute), *Tüketici Fiyat Endeksi, Ağustos 2026*, news bulletin of 3 September 2026: CPI +31.51% against August 2025 (monthly +1.84%). Listed on [tuik.gov.tr](https://tuik.gov.tr) ("%31,51 (yıl)") and published at [veriportali.tuik.gov.tr/tr/press/58290](https://veriportali.tuik.gov.tr/tr/press/58290); verified there on 6 September 2026.

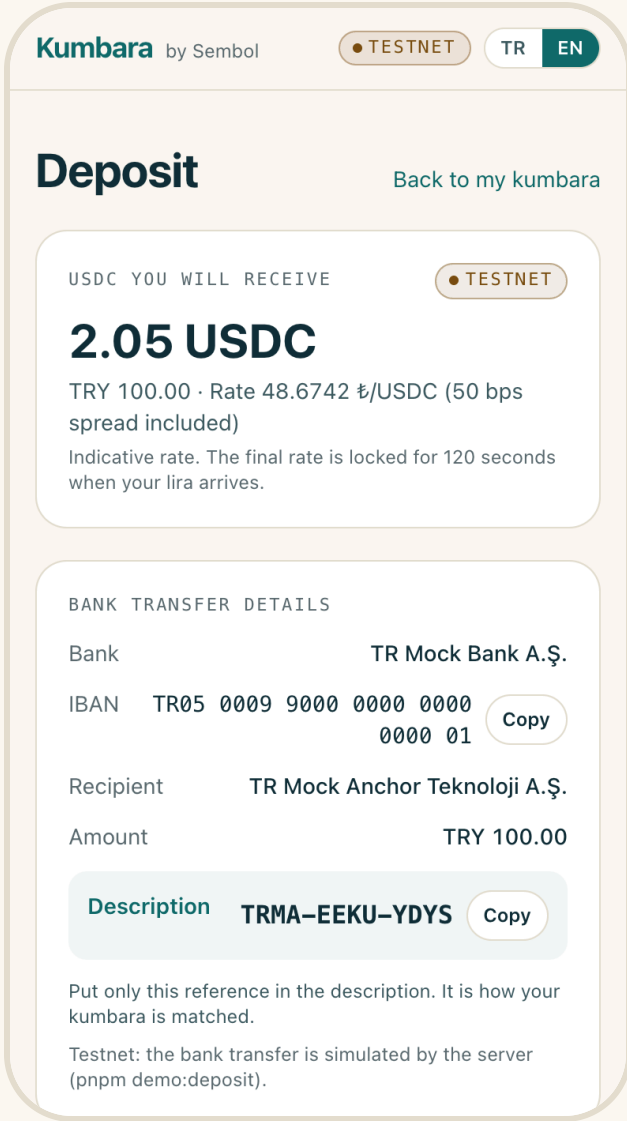
# Four screens. Tap to account: ~10 s. No seed phrase, no XLM, no install.



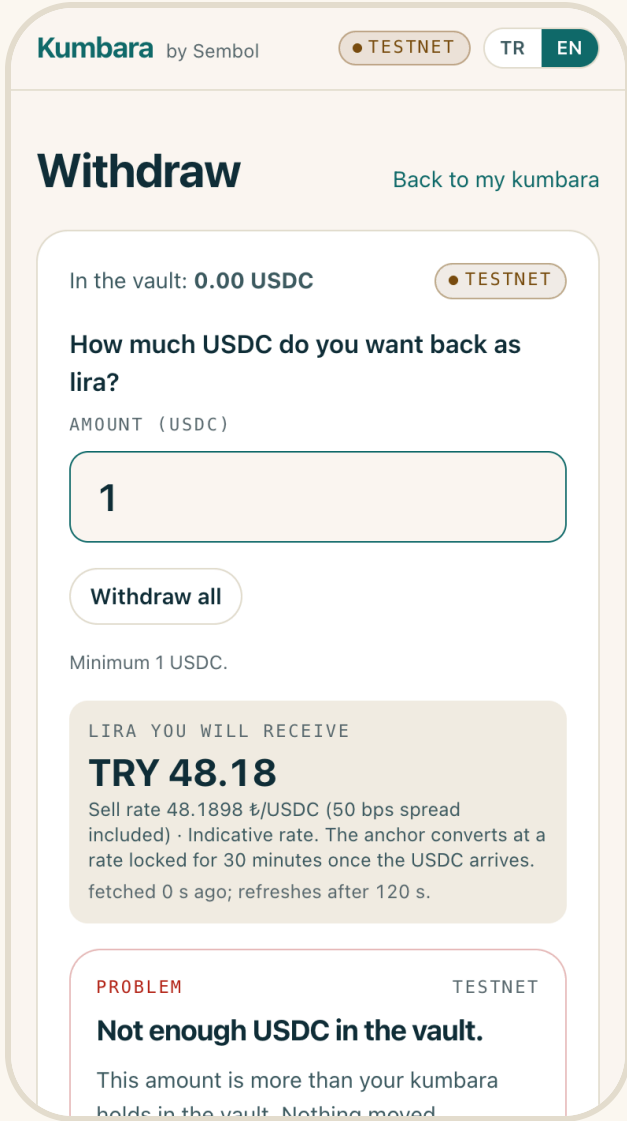
**Onboard.** One button; Face ID creates the key and the relay deploys an OpenZeppelin smart account. Measured 8–16 s tap to Savings in browser runs.



**Savings.** USDC in the DeFindex vault with its lira equivalent (Reflector rate), spending limit, recovery. Every figure carries the TESTNET badge.



**Deposit.** IBAN, recipient and a reference code, exactly like an online bank transfer; the anchor converts lira to USDC.



**Withdraw.** A sell quote in lira; two Face ID approvals; the anchor pays the visitor's own IBAN.

# Seven steps, one flow: anchor in front, DeFindex behind, a fee-sponsoring relay underneath.

- 1

The visitor opens an OpenZeppelin smart account with Face ID; the relay pays the fees.
- 2

A spending limit (1,000 USDC per transaction) is written on-chain in the background.
- 3

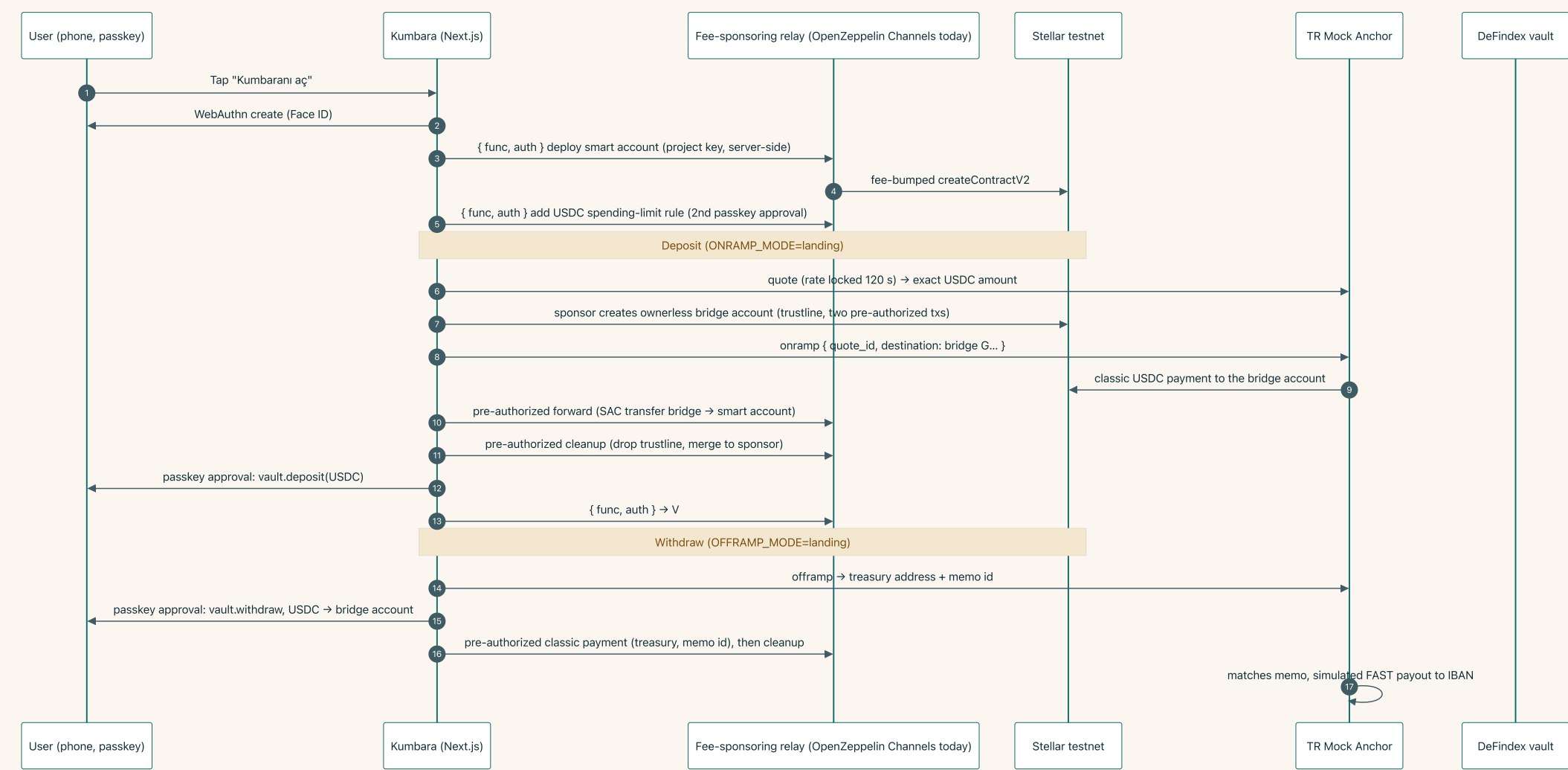
The anchor issues an IBAN and a reference; the visitor sends lira (simulated by the server on testnet).
- 4

The anchor converts the lira to USDC and pays an ownerless, pre-authorized bridge account.
- 5

The bridge account forwards the USDC to the kumbara and closes itself; nobody holds its key.
- 6

One Face ID puts the USDC into the DeFindex vault (no yield on testnet).
- 7

A withdrawal runs the path backwards: vault → bridge account → anchor → lira to the IBAN.



# Anchors pay classic accounts. Smart accounts are contracts. Ramp payouts could not reach them.

## The finding

The anchor's on-ramp rejects a contract (C...) destination and its watcher matches classic payments only: a payout cannot land in a smart account, and a smart account's Soroban transfer is never matched on the way back.

## The solution: ownerless, pre-authorized bridge accounts

A throwaway classic account whose only valid transactions are signed before it exists: the **forward** at sequence +1 (USDC to the user's smart account, the exact quoted amount) and the **cleanup** at sequence +2 (trustline dropped, reserves back to the sponsor). The master key is wiped in-process: **no key can sign anything else**. In reverse, the same primitive carries withdrawals to the anchor's treasury with the memo id.

10 / 10

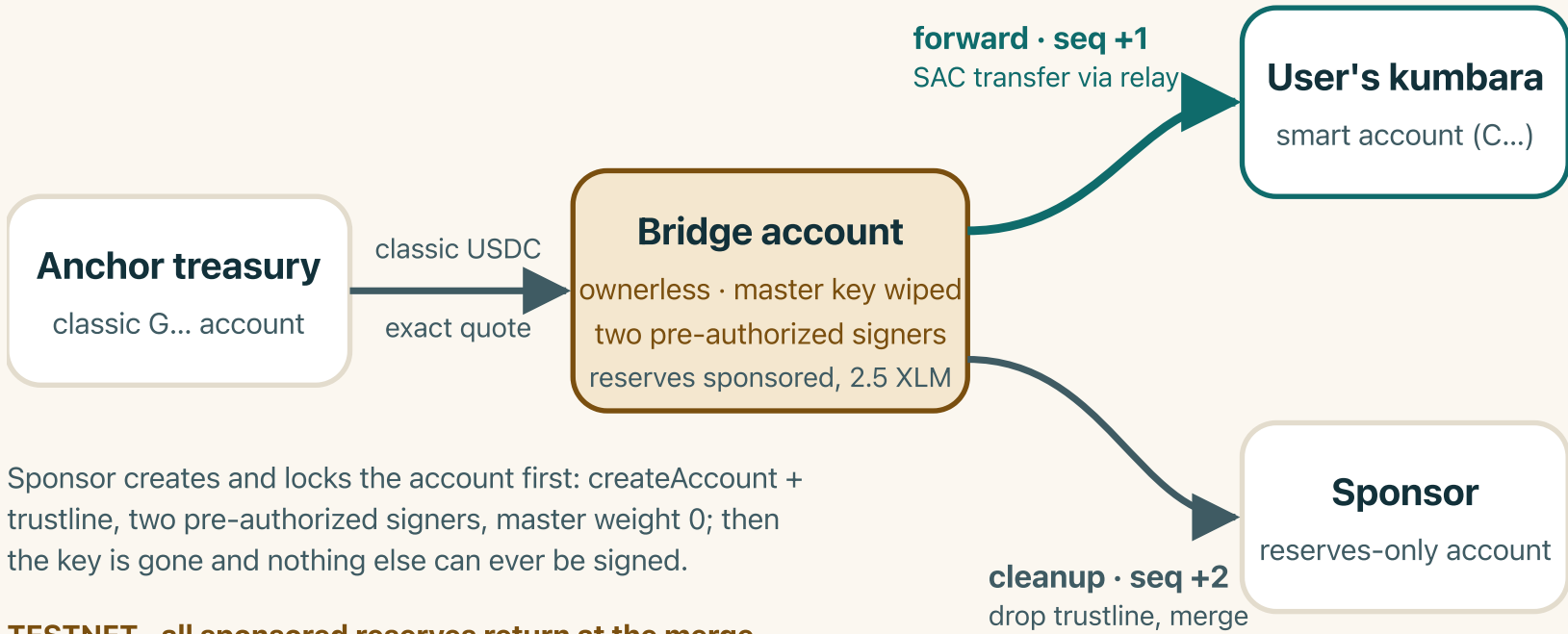
exact amount matches in ten scripted deposits

900

stroops net per deposit for the sponsor; reserves return at cleanup

4

signatures the bridge key ever produces, asserted by a unit test



Sponsor creates and locks the account first: createAccount + trustline, two pre-authorized signers, master weight 0; then the key is gone and nothing else can ever be signed.

**TESTNET · all sponsored reserves return at the merge**

| EVIDENCE (STELLAR TESTNET, STELLAR.EXPERT/EXPLORER/TESTNET/TX/...) | TRANSACTION         |
|--------------------------------------------------------------------|---------------------|
| Anchor pays the bridge account (classic payment)                   | 778215b3...e3c3a8f1 |
| Pre-authorized forward: USDC lands in the smart account            | f580e030...f22c9eab |
| Pre-authorized cleanup: trustline removed, account merged          | 6223c61c...7416291d |

Reported to the anchor team with the two fixes that would retire the bridge: SAC delivery to contract wallets and Soroban-aware matching on the way back. Full hashes in docs/anchor-notes.md.

# Listed building blocks, wired end to end. Kumbara writes no wallet, vault or strategy code.

| BUILDING BLOCK                                      | WHAT IT DOES IN THIS APP                                                                                                                                                                                    | ADDRESS (TESTNET, STELLAR.EXPERT)                   |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| OpenZeppelin smart accounts + spending-limit policy | Every kumbara is an audited smart account whose only signer is the visitor's passkey; a USDC-scoped rule caps 1,000 USDC per transaction on-chain. Via @sembol/passkey-react and stellar/smart-account-kit. | WASM 1b5f4534...97785a · policy CABXBYJN...2TIP5G   |
| DeFindex vault + hodl strategy                      | Deposits go through the vault's real invest path into its strategy and withdrawals unwind from it. The strategy is DeFindex's own hodl crate, built unmodified; no yield on testnet, none promised.         | vault CAT76PQM...MKSKL · strategy CDFLOE4H...JLDMQN |
| TR Mock Anchor · SEP-1/6/38 + Partner API           | The regulated lira side: IBAN and reference, quotes with a 120 s lock, on-ramps to the bridge account, off-ramps by memo, simulated FAST payouts. Issuer and endpoints read from its stellar.toml.          | USDC SAC CBIELTK6...XQDAMA                          |
| Fee-sponsoring relay (OpenZeppelin Channels today)  | Signed authorization entries go to our server route, which adds the project key and forwards to the relay; the visitor never sees XLM. Per-IP and per-booth caps, sponsor threshold.                        | channels.openzeppelin.com/testnet                   |
| Reflector                                           | The USD/TRY foreign-exchange oracle behind the lira equivalent on the Savings and stats pages (read by simulation, anchor rate as fallback).                                                                | CBKGPWGK...4DXMCJZC (mainnet oracle, read only)     |

**Upstream:** "Fix wallet creation crash on default Next.js apps" was merged into [stellar/smart-account-kit](#) on 19 August 2026 and shipped in 0.6.2 · [github.com/stellar/smart-account-kit/pull/4](#)

**Open-source plan:** the repository is MIT-licensed. Kumbara deploys no proprietary contracts: the smart account, WebAuthn verifier and spending-limit policy are OpenZeppelin's audited contracts, the vault and strategy are DeFindex's; the bridge-account logic (`lib/landing`) is MIT in the repo with its tests, and the repository goes public at award time.

# Everything on this slide is public, linkable and re-runnable.

Kumbara by Sembol

TESTNET

TR

EN

Kumbara live

live updated 1:17:47 AM

Public, read-only. Testnet: no real lira moves.

Since event start

All time

1

kumbaras opened

TESTNET

1

deposits completed

TESTNET

0

withdrawals completed

TESTNET

TRY 100.00

TRY in

TESTNET

TRY 0.00

TRY out

TESTNET

26.79

USDC

USDC in the vault (live)

TESTNET

**kumbara.sembol.xyz/stats**: headline numbers with a TESTNET badge, live feed, onboarding curve, timings. The *since event start* / *all time* toggle counts from 19 September 08:00 and shows all-time until then; seed and E2E accounts are excluded by default.

6

real-browser round trips in CI: onboard, deposit, withdraw, booth, stats, failure screens · every push and every 6 h against production

25

failure states, each a TR/EN screen with one action; every flow resumes from stored state

94 · 100 ·  
100 · 100

Lighthouse mobile on /stats: performance, accessibility, best practices, SEO

E2E · passing · latest production run 6  
Sep 2026

GitHub Actions, workflow e2e.yml (repository badge available once the repo is public)

ON-CHAIN EVIDENCE (STELLAR TESTNET)

TRANSACTION

Vault deployed through the DeFindex factory with the hodl strategy attached

dbf0d5b7...6faa8f96

Deposit with invest=true moved USDC into the strategy (invest)

b615e7dc...76b7f981

Withdrawal unwound the shares from the strategy (divest)

466f48d2...07a9f5ca

Recorded fallback demo: docs/demo/round-trip.mp4, the full round trip on kumbara.sembol.xyz, 2:39 with a caption per step.

# Non-custodial software in front of a licensed anchor.

- **Kumbara never holds lira, USDC or keys.** Every transfer is authorized by the visitor's own passkey on their own smart account; the bridge accounts have no owner.
- **The licensed anchor is the regulated party:** it holds the fiat, performs KYC and pays out only to the customer's own named IBAN.
- **No payments feature in Türkiye:** no send, no P2P, no merchant flow. Deposit into a vault and withdraw back to your own IBAN, nothing else.
- **Turkish AML rules (MASAK) shape the UX:** named-IBAN in and out, an on-chain per-transaction cap, anchor-side limits and no cash or third-party paths.
- **A Turkish regulatory opinion comes first, self-funded** (legal fees are not an eligible SCF cost); it is obtained during Tranche 1, and the mainnet path stays behind a capped, presenter-only flag until the opinion and the audit are in.

# One builder, already shipping in the ecosystem.

## Ahmed

Founder · builds and pitches

Instaward grantee (Stellar Türkiye). Author of [@sembol/passkey-react](#), the passkey wallet layer Kumbara runs on, and contributor to [stellar/smart-account-kit](#). Designed and built Kumbara end to end: the app, the bridge accounts, the CI round trips and the evidence in this deck.

Site [ahmed.bet](#)

LinkedIn [linkedin.com/in/ahmed-zm](#)

GitHub [github.com/keyboard01](#)

### SHIPPED UPSTREAM

**stellar/smart-account-kit #4**: "Fix wallet creation crash on default Next.js apps", merged 19 August 2026, in release 0.6.2.

### PUBLISHED

**@sembol/passkey-react** on npm: passkey creation and signing, recovery, spending-limit policy UI. Every screen in this deck runs on it.

### FUNDED

**Instawards grantee**. Kumbara is the consumer proof for that work, built for this submission.

Groundwork, then testnet-complete and secured, then mainnet. Every line has a cost and a check.

| DELIVERABLE                                                                                                                          | HOW REVIEWERS VERIFY                             | USD                 |
|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|---------------------|
| TRANCHE 1 · GROUNDWORK · ≤ 60 DAYS                                                                                                   |                                                  | SUBTOTAL [\$14,000] |
| Anchor shortlist evaluated against SEP readiness (1, 6, 10, 12, 24, 38); one selected; LOI signed or a documented go/no-go           | evaluation doc, LOI or go/no-go memo in the repo | [\$3,000]           |
| Audit Bank application submitted with scope: policy install, bridge accounts, server path                                            | submission confirmation and scope document       | [\$1,000]           |
| Relay budgets and allowlists: per-project caps, contract allowlist, refusal alerts                                                   | config in repo; a refused call's log and tx      | [\$6,000]           |
| Hosted (Vercel + Turso) and self-host (Fly + SQLite) deployment docs with scripted setup                                             | docs URLs; a fresh deploy from the docs          | [\$4,000]           |
| TRANCHE 2 · TESTNET-COMPLETE AND SECURED · ≤ 90 DAYS                                                                                 |                                                  | SUBTOTAL [\$31,000] |
| Threat model in SDF's STRIDE template format (assets, threat table, mitigations)                                                     | docs/security/threat-model.md                    | [\$5,000]           |
| On-chain monitoring plan in the builders template, implemented for the vault, policy, sponsor and bridge accounts with alert routing | monitoring-plan.md; a live alert on testnet      | [\$8,000]           |
| Audit of the policy install, bridge accounts and server path via Audit Bank (our scoping and engineering support only)               | audit report link                                | [\$3,000]           |
| Findings remediated, with regression tests                                                                                           | fix commits, CI green, auditor re-check          | [\$8,000]           |
| Mainnet path behind a capped, presenter-only flag: real USDC, enforced budgets                                                       | mainnet tx hashes from the capped path           | [\$7,000]           |
| TRANCHE 3 · MAINNET LAUNCH AND PUBLIC PILOT · ≤ 90 DAYS                                                                              |                                                  | SUBTOTAL [\$32,000] |
| Mainnet launch: vault, strategy and policy contracts registered; sponsor account and relay budgets live                              | mainnet contract IDs on stellar.expert           | [\$9,000]           |
| Public pilot cohort on the selected anchor's production rails; validation-focused user testing (within the 1/3 exception)            | /stats on mainnet, tx hashes, cohort report      | [\$12,000]          |
| Metric window: 30-day NAV measurement, public dashboard, footprint registered                                                        | NAV dashboard URL; registered IDs                | [\$3,000]           |
| Minor: native mobile (Expo) SDK for the passkey layer, published with docs                                                           | npm package and docs URL                         | [\$8,000]           |
| Total ask: the sum of the bracketed lines (Integration Track "Medium" band, \$50k–100k)                                              |                                                  | [\$77,000]          |

FINAL-TRANCHE METRIC COMMITMENT

|                                              |                                                                                                                                                       |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Metric type                                  | DeFindex vault NAV from TRY deposits (USDC in Kumbara's vault position, mainnet)                                                                      |
| Threshold                                    | [X] USDC                                                                                                                                              |
| Measurement window                           | 30 days after the pilot launch, measured over the window, not a snapshot                                                                              |
| On-chain footprint, registered at award time | vault contract, strategy contract, spending-limit policy contract, sponsor account (sponsored bridge accounts per CAP-33), app-created smart accounts |

The panel ratifies or adjusts the threshold during panel review. Self-generated, wash or gamed activity is excluded; seed and E2E accounts are already tagged out of every number.

SELF-FUNDED · NOT IN THE ASK

Turkish regulatory opinion (legal fees), entity setup (registration costs), the audit fee itself (Audit Bank), and any marketing beyond the validation-focused user testing allowed in tranches 2 and 3. Mainnet waits for the opinion and the audit regardless.

Expansion filter, after the pilot: high inflation plus an existing licensed Stellar anchor. Same app, same bridge, a different anchor in front.

USD cells are bracketed placeholders for Ahmed; subtotals and the total are their sums. Paid in XLM in tranches of 10% on acceptance, 20% at MVP, 30% at testnet, 40% at mainnet: SCF Handbook, Build Award, "Budget & Deliverable Guidelines" → "How Tranche Payments Work" (stellar.gitbook.io/scf-handbook/scf-awards/build-award/budget-and-deliverable-guidelines). Ineligible per the same page: audit costs, marketing beyond the Integration Track exception, legal fees, entity registration.